

Jie Du

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Education

Houston Community College
Electrical Engineering | 4.0 GPA

Houston, TX
August 2024 – May 2026

Awards

Distinguished Honor Graduate, Advanced Individual Training (AIT), U.S. Army

Professional Experience

Signal Support Specialist

June 2023 – current

United States Army Reserve

Houston, TX

- Engineered and administered communication networks, configuring satellite systems, tactical radios, and network devices to ensure mission-critical connectivity by implementing encryption protocols to secure data transmissions.
- Troubleshooted and resolved complex hardware and software issues across diverse operational environments, including desert, rainforest, and mountainous terrains, to maintain 100% system readiness.
- Applied cybersecurity principles to protect Army networks from potential threats, configuring system environments and frequencies to fortify communications against interception and interference.

Research Experience

Undergraduate Research Assistant

May 2026 – current

National Center for Airborne Laser Mapping (NCALM), University of Houston

Houston, TX

- Processed and analyzed airborne LiDAR point cloud data using industry-standard geospatial tools, extracting meaningful terrain features from large-scale datasets collected across diverse environments.
- Applied machine learning techniques to automate terrain and ground classification within LiDAR point cloud datasets, improving accuracy and efficiency of surface model generation.

Undergraduate Research Apprentice

August 2025 – November 2025

University of Los Angeles

Los Angeles, CA

- Secured a competitive research position with Professor Dejan Markovic to contribute to cutting-edge research on FPGA-based deep learning accelerators.
- Designed and implemented custom hardware accelerators on FPGAs using Verilog and Xilinx Vivado, engineering efficient parallel architectures for convolutional neural network operations.
- Optimized neural network models for hardware deployment via quantization and pruning using PyTorch and C/C++ to accelerate inference performance on custom FPGA hardware.

Project Experience

Full-Stack Mobile App Development

Congressional App Challenge

Saipan, CNMI

- Architected and developed a full-stack mobile application for iOS using Swift and SwiftUI, implementing a scalable Firebase backend for real-time data storage and user authentication.
- Managed the complete project lifecycle including technical documentation outlining the app's architecture, functionality, and community impact.

Technical Skills

Programming Languages: Python, JavaScript, Swift, SwiftUI, HTML, C/C++

Security Clearance: Secret

Cybersecurity & Network: Radio and Satellite Systems, FPGA, Army Rifles, Network Defense, Encryption Protocols

Leadership Skillsets: Curriculum Development, Team Management, Engineering Lifecycle Management, Mentorship